

Students from a local school carried out a survey on the Afon Hwch, a river that flows into Llyn Padarn, during July of 2009. The site of the survey is shown on the map on page 14. The results are recorded below:

Number of organisms						
stonefly nymph	mayfly nymph	caddis fly larvae	blood worm	leeches	sludge worms	rat tailed maggot
3	17	7	0	5	0	0

The following table can be used to assess the condition of the water.

Organism present	Levels of pollution	Oxygen concentration
no visible life	very high	no oxygen
rat tailed maggot	high	low
sludge worm		
leeches	moderate	moderate
blood worm		
caddis fly larvae	low	moderate to high
mayfly nymph	none	high
stonefly nymph		

- (c) (i) State the scientific term which is used to describe organisms that are used to detect levels of pollution. [1]

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- (ii) Consider the information provided to describe the water conditions in the Afon Hwch. [2]

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Examiner
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- (iii) It was recommended that further scientific studies were needed to show that the algal blooms were not caused by fertilisers used on the land around the lake. Describe how students could extend their survey to test this idea. [2]

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